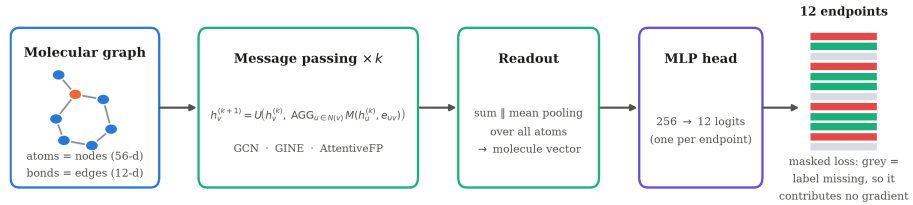
Applied Machine Learning / Deep Learning · Group Project 8

# Molecular Toxicity Prediction

## Phase 2: Graph Neural Networks on Tox21


| | |
|---|---|
| Pankaj Upadhyay | 2401010317 |
| Prakhar Rawat | 2401020045 |
| Om Chimurkar | 2401010306 |
| Harshit Kudhial | 2401010182 |
| Vishuti Jamwal | 2401010506 |


Atoms as nodes · bonds as edges · learned features

github.com/pankaj-cod/AML_DL-Project

# 1 The Question We Set Out to Answer

## Phase 1 gave us a strong baseline




ROC-AUC on chemical families the model has **never seen**

## …but fingerprints are frozen

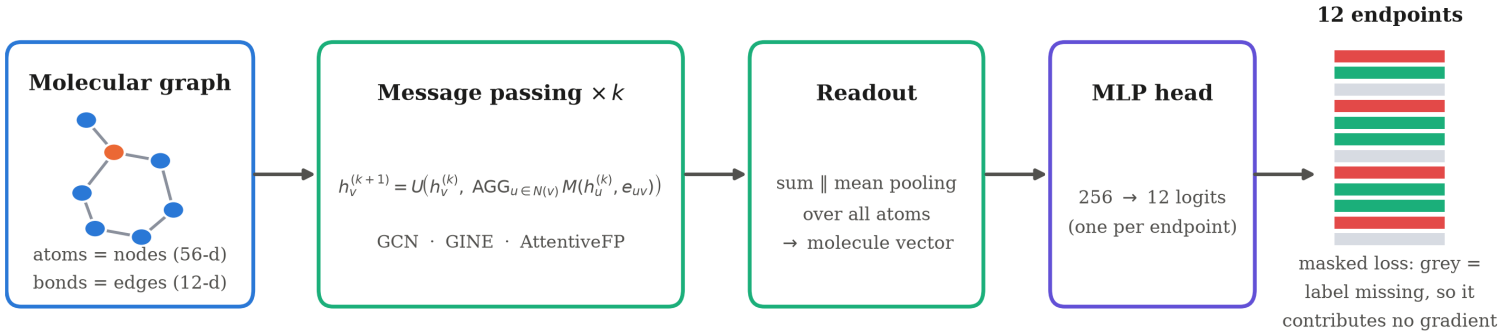
- A fixed checklist of substructures — the **same** features for all 12 endpoints.
- Chosen by a rule, never by the task.
- So: could a model **learn** better features itself?

**PHASE 2 QUESTION**

Can a neural network that reads the molecule as a **graph of atoms and bonds** beat hand-engineered fingerprints?

Tested where it matters: on chemistry it has never seen before.

# 2 How the Network Reads a Molecule




## 1. Build the graph

Every **atom** is a node with 56 features (element, charge, ring…). Every **bond** is an edge with 12.

## 2. Pass messages

Each atom mixes in its neighbours' information. Repeat **4 times**, so every atom “sees” 4 bonds around it.

## 3. Pool and predict

Combine all atoms into one vector for the molecule, then output **12 toxicity scores** at once.

# 3 What We Trained

## GCN

**0.817**
ROC-AUC

Averages its neighbours. Simplest version — ignores bond type.

multitask: 1 network → 12 endpoints

## GINE-MPNN

**0.803**
ROC-AUC

Sums messages that **include the bond**. Most expressive.

multitask: 1 network → 12 endpoints

## AttentiveFP

**0.818**
ROC-AUC

Uses **attention**: some neighbours count more than others.

multitask: 1 network → 12 endpoints

## Plus three controls, so the comparison is honest

### FP-MLP 0.814

deep net on the **same fingerprints** — separates “deep” from “graph”

### Hybrid 0.809

graph + fingerprint inside one network, trained together

### Ensemble 0.844

average the **rankings** of Random Forest and the best GNN

# 4 Two Problems We Had to Solve

## 1. 17% of labels are missing

Only **3,019 of 7,686** molecules have all 12 labels.

✗ **Drop them** — loses 61% of the data

✗ **Fill with 0** — invents false “non-toxic” labels

✓ **Mask them out** — use every real label, invent nothing

$$\text{loss} = \Sigma\ \text{mask} \cdot \text{BCE} \ / \ \Sigma\ \text{mask}$$

## 2. Very few molecules are toxic

Just **2.8%** on NR-PPAR-gamma. Always answering “non-toxic” is **97% accurate** and completely useless.

- Weight each toxic molecule **≈34×** more in the loss.
- Report **ROC-AUC** and **PR-AUC** — never accuracy.

**WHY THIS IS THE DEEP-LEARNING PART**

One network predicts all 12 endpoints together. The **mask** lets it learn from every molecule even when most of its labels are missing — something 12 separate classical models cannot do.

# 5 Results on Unseen Chemistry

Mean over 12 endpoints, 3 scaffold splits

| Model | ROC-AUC | PR-AUC |
|---|---|---|
| **Random Forest** | **0.834** | **0.496** |
| XGBoost | 0.816 | 0.475 |
| FP-MLP | 0.814 | 0.440 |
| GCN | 0.817 | 0.450 |
| GINE-MPNN | 0.803 | 0.399 |
| AttentiveFP | 0.818 | 0.436 |
| Hybrid | 0.809 | 0.398 |
| **Ensemble RF + GNN** | **0.844** | **0.503** |

## The graph networks tie — they don't win

GNNs reach **0.803–0.818**, level with XGBoost (0.816) and the FP-MLP (0.814). Those differences are **not significant** ($p > 0.3$).
But Random Forest (0.834) beats every GNN — and that difference **is** significant ($p = 0.004$).

## Combining the two wins

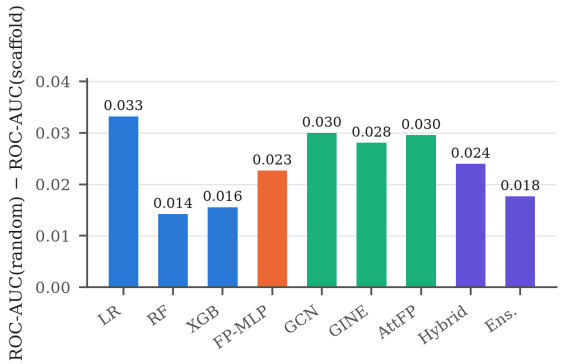
**0.844**
Ensemble ROC-AUC

**0.503**
Ensemble PR-AUC

Validation gives the GNN **0.46** of the weight — proof it sees something the fingerprint misses. Beats RF on 26 of 36 pairs ($p = 0.001$).

# 6 New Chemistry — and Why GNNs Didn't Win

## Accuracy lost on unseen scaffolds




## The graph models lose about twice as much

- Random Forest drops **0.014**; GCN and AttentiveFP drop **0.030**.
- On *random* splits the GNNs tie Random Forest (0.847 vs 0.849) — the gap appears only on new chemistry.
- For truly novel molecules (similarity < 0.3) everything falls to about 0.64–0.72.

## Too little data

~6,100 training molecules. Learned features need more than that.

## A very strong baseline

The 217 descriptors matter: without them XGBoost falls to 0.751.

## The control proves it

FP-MLP — deep learning on the old features — also ties, at 0.814.

# 7 Conclusion

## 0.834

### Fingerprints still win

Random Forest beats every graph network on unseen scaffolds ($p = 0.004$).

## 0.817

### GNNs match XGBoost

GCN 0.817 vs XGBoost 0.816 — no significant difference.

## 0.844

### Together is best

RF + GNN beats either alone: they make different mistakes.

**TAKEAWAY**

On a dataset this size, **combine** engineered and learned features — don't replace one with the other. And always test on scaffold splits, or the numbers flatter you.

Next: pre-trained graph networks · a GNN hyper-parameter search · an applicability-domain warning.